PREOPERATIVE DIAGNOSIS: Severe end-stage tricompartmental osteoarthritis of the left knee, refractory to non-operative modalities including physical therapy, unloader bracing, and corticosteroid/hyaluronic acid intra-articular injections.

POSTOPERATIVE DIAGNOSIS: Severe tricompartmental osteoarthritis, left knee.

PROCEDURE: Left total knee arthroplasty using Zimmer NexGen posterior-stabilized cemented modular prosthesis.

SURGICAL TECHNIQUE: The patient was brought to the operating suite, positioned supine on the operating table, and general endotracheal anesthesia was induced. A high-thigh pneumatic tourniquet was placed on the left lower extremity. After thorough betadine skin scrub and alcohol prep, the extremity was draped in a sterile orthopedic fashion. The limb was exsanguinated with an Esmarch bandage, and the tourniquet was inflated to 250 mmHg.

A standard anterior midline longitudinal skin incision of approximately 14 cm was created over the left knee, followed by a medial parapatellar arthrotomy. The patella was gently everted laterally, and the knee was brought into maximal flexion. Extensive denudation of articular cartilage with large osteophytes was identified along the medial and lateral femoral condyles, intercondylar notch, and tibial plateau. The anterior cruciate ligament was chronically absent, and degenerate medial and lateral meniscal remnants were cleanly excised.

An intramedullary femoral alignment guide was positioned, and distal femoral resection was carried out at 5 degrees of valgus. Anterior, posterior, and chamfer cuts were completed using the anterior-referencing 4-in-1 cutting block with optimal rotational alignment parallel to the transepicondylar axis. An extramedullary tibial alignment jig was assembled, perpendicular to the mechanical axis of the tibia with 3 degrees of posterior slope, and the proximal tibial plateau was resected removing 9 mm of bone from the unaffected side. Soft tissue balancing was performed; a tight medial flexion gap was addressed with judicious subperiosteal release of the deep medial collateral ligament.

Trial femoral, tibial, and patellar components were placed along with a 10 mm trial articular spacer. The knee demonstrated complete extension to zero degrees, flexion to 125 degrees, and symmetric 1 mm coronal stability in both flexion and extension without patellar tilt or subluxation (no-thumb test negative). The bone surfaces were prepared using pulsatile jet lavage and dried. High-viscosity antibiotic-impregnated polymethylmethacrylate bone cement was mixed and pressurized into the cancellous bone beds of the distal femur and proximal tibia. The final prosthetic components were seated under firm axial compression. Extruded excess cement was carefully debrided. The tourniquet was deflated; meticulous hemostasis was secured with electrocautery. Layered closure was executed using #1 Vicryl for the arthrotomy, 2-0 Vicryl for subcutaneous tissues, and running 3-0 Monocryl with topical Dermabond mesh for the skin. Sterile dressings and an articulated knee immobilizer locked in extension were applied.